

A Survey of Music Recommendation Systems

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ABSTRACT

The contradiction between users' music consumption and the abundance of music data has spurred the emergence of music recommendation systems. These systems merge recommendation systems with music suggestions, serving as a widely utilized solution in real-world scenarios and presenting a multifaceted challenge. Many methods of recommendation systems need to combine the characteristics of music recommendation itself. For example, with the rise of mobile Internet and the popularity of smartphones, the places where users listen to music are not limited by time or space; the emotions of users when listening to music also have a great influence on music recommendations, so it is necessary to study personalized music recommendation systems. In this review, we mainly focus on deep learning-based music recommendation systems, starting from the perspectives of neural network structure and personalized recommendation systems, discussing the principles, characteristics, and applications of different models. We also review the main methods and latest research results used in music recommendation systems. This review should be of great help to scholars who are new to the field of music recommendation or want to quickly grasp the recent advances in the field.

CCS CONCEPTS

• **Information systems** → Information retrieval; Retrieval tasks and goals.

KEYWORDS

Music recommendation system, Collaborative filtering, Deep neural network, Artificial intelligence

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1 INTRODUCTION

Recommendation systems are used to help users find desired information from massive data, which can include music, products, movies, books, news, websites, courses, and more. In 1992, researchers including David Goldberg from Xerox Corporation first applied recommendation systems based on collaborative filtering algorithms^[1]. In 2012, AlexNet^[2] achieved first place in the ImageNet competition, and subsequently, deep learning has been widely applied in various domains such as natural language understanding, image recognition, and speech processing, including in recommendation systems.

Music recommendation is particularly noteworthy because it possesses several notable characteristics that differentiate it from other types of recommendation systems. For instance, songs are generally short, typically only a few minutes long, and there is a huge number of songs, with libraries containing tens of millions of songs. An example of a music recommendation application is Spotify, which has over 150 million active users^[3] and an extensive song library numbering in the tens of millions. NetEase Cloud Music is a prominent entity in China's music streaming industry, launching on April 23, 2013, and quickly becoming popular nationwide. According to a recent evaluation, this service had around 800 million users by 2019, with an estimated market value of approximately 9 billion US dollars^[4].

Therefore, in this review, based on the characteristics of music recommendation itself and the latest research progress, we mainly discuss different music recommendation system models from two different perspectives: the structure of neural networks and personalized recommendation systems. The following two subsections respectively introduce the characteristics of music recommendation and the contributions and overall structure of this paper.

1.1 Characteristics of music recommendation

Music songs are short: Music songs are generally short in duration, which means that users spend short amounts of time consuming music. Evaluations of music can be made within a short period, and users often know whether they like a song or not after listening to only half or even the beginning of a song.

Different abstraction levels: Music recommendation can operate at various abstraction levels. Music recommendation systems can suggest music based on audio features, artists, albums, and individual songs, enabling recommendations grounded in the music's content itself.

Repeat listening: A piece of music can be listened to repeatedly by users, unlike movies which are generally only watched once. This implies that users do not mind recommendations for music they already like.

Listening in sequence: Unlike movies and books, music is frequently listened to in a particular order. Consequently, the music recommendation issue can be viewed as a sequential recommendation problem for music playlists.

Each user has their own music taste, and how they choose music depends on their surroundings and mood. Therefore, emotion-based recommendation is also one of the research directions.

1.2 Contributions and structure of this survey

This paper makes the following contributions: First, it addresses a gap in the literature by focusing on the latest research from the past two years, particularly examining the impact of generative artificial intelligence and large models on music recommendation. This paper organizes the recent research results from the past two years. Second, some recent reviews have been too general about the use of deep learning in music recommendation and have not classified the use of neural networks well. Our work categorizes music systems using deep learning methods based on recent main network structures. Third, prior reviews have solely concentrated on either content-based personalized music recommendation or deep learning-based music recommendation methods, whereas this paper provides a comprehensive summary of both approaches.

The paper's structure is outlined as follows: The first part provides an introduction to the research background of music recommendation systems and the unique traits of music recommendation. Following that, the subsequent section delves into various approaches for music recommendation, encompassing collaborative filtering techniques and deep learning-based methodologies. The third section introduces common evaluation methods for music recommendation systems, including offline evaluation, online evaluation, and user surveys. The fourth section discusses publicly available datasets in the field of music recommendation systems. The fifth section introduces the challenges faced by music recommendation systems. The sixth section provides a summary and discusses future directions for music recommendation systems.

2 MUSIC RECOMMENDATION METHODS

2.1 Collaborative filtering

Within recommendation systems, collaborative filtering stands out as the predominant method employed. In 1992, researchers including David Goldberg from Xerox Corporation first applied recommendation systems based on collaborative filtering algorithms^[1]. Collaborative filtering was initially introduced in the Tapestry email system, where users could annotate emails, and these annotations could be used for collaborative filtering. In 2003, a paper published by Amazon^[5] made collaborative filtering a hot topic and a foundational recommendation model for future research. The idea behind collaborative filtering revolves around suggesting items favored by users with similar preferences to the target user, or proposing akin items to those already favored by the target user. The crucial aspect lies in computing the resemblance between users or items. Similar to other recommendation systems, collaborative filtering finds

extensive application in music recommendation, predominantly categorized into user-based and item-based recommendations.

Collaborative filtering algorithms operate on user-to-music interaction data, encompassing explicit data like user ratings for music and implicit user feedback such as play counts and skipped tracks. A technique employing play coefficients was introduced in^[6], utilizing implicit user feedback to mitigate the data sparsity issue encountered in recommendation systems when information availability is limited. In^[7], collaborative filtering algorithms based on nearest neighbor classifiers were amalgamated to develop a hybrid recommendation system, offering personalized recommendation services to users. Furthermore, the paper discussed alternative approaches like content-based filtering, collaborative filtering, singular value decomposition (SVD), and factorization machines (FM). Ultimately, the proposed system was integrated using SVD and FM.

Reference^[8] is a review of the use of collaborative filtering techniques in music recommendation systems, with a specific emphasis on investigating two approaches: user-based and item-based recommendations. Various metrics for gauging the similarity between users and items are explored, including Euclidean distance, cosine similarity, Pearson correlation coefficient, among others. Moreover, it conducts a comparative assessment of diverse evaluation metrics that gauge the efficacy of recommendation systems. For more information on collaborative filtering, refer to references^{[9], [10], [11], [12]}.

2.1.1 Limitations of collaborative filtering. Both user-based and item-based collaborative filtering methods do not utilize the content of items themselves, thus they are not constrained by the content of items. This made them popular in the early stages of recommendation systems, but they also have limitations. Firstly, the assumption that users with similar behavior have similar music preferences has not been widely researched. Secondly, the number of comments users make on songs is relatively small compared to the number of songs in the library, leading to the problem of data sparsity, which becomes more pronounced as the dataset grows larger. Collaborative filtering algorithms also suffer from algorithm bias, where more popular music tends to receive more clicks, while long-tail music is rarely listened to. Recommending popular music to users is relatively safe, but it leads to algorithm bias as users do not experience surprises. The cold start problem arises in the early stages of the recommendation system when there is almost no user feedback, resulting in poor recommendation performance.

2.2 Deep learning-based methods

Similar to the domains of computer vision and natural language processing, deep learning techniques have seen a surge in their utilization within recommendation systems in recent years. Deep neural networks are predominantly leveraged in music recommendation for feature extraction, encompassing tasks like extracting implicit features from music acoustic signals and capturing temporal patterns from user playlists. Notable network architectures utilized in music recommendation encompass convolutional neural networks, recurrent neural networks, attention mechanisms, Transformer models, graph neural networks, and others.

2.2.1 Convolutional neural networks. Convolutional Neural Networks (CNNs) have demonstrated remarkable efficacy in domains including speech recognition, image recognition, and natural language processing. They have advantages such as translation invariance, parameter sharing, and hierarchical structure, which enable them to effectively extract features from data. However, CNNs are sensitive to the size and shape of input images, requiring preprocessing to ensure consistency in input data, which adds complexity to their usage. CNNs typically consist of convolutional operations, pooling operations, and activation functions.

Most recommendation systems use collaborative filtering methods, but these methods are plagued by the cold start problem; they are ineffective at recommending new songs and unpopular songs when user data is insufficient. Considering these issues, utilizing convolutional neural networks to extract insights from available data and employ content-based approaches emerges as a viable solution. Reference^[13] proposed using latent factor models for recommendation to mitigate the cold start issue. In instances where user data is unavailable, latent factors are predicted from music audio. Training a convolutional neural network (CNN) to forecast user-item latent variables from collaborative filtering matrix factorization data was also suggested. Subsequently, this CNN could independently deduce these latent representations solely from audio signals. Reference^[14] introduced a novel deep neural network architecture tailored for content-based music recommendation, dubbed Music CNNs. The researchers compiled a dataset comprising more than 600,000 songs and partitioned each song into roughly 20 music segments. To leverage the capabilities of CNNs, they transformed music segments into “images” using Fourier transform and input them into the CNN. Experimental results showed that Music CNNs generally provide more accurate recommendations compared to the benchmark methods.

Reference^[15] proposed a personalized music recommendation system (PMRS) integrating CNNs with collaborative filtering algorithms. The system employs convolutional neural networks to categorize music genres based on audio signals, while the collaborative filtering algorithm furnishes music recommendations to users utilizing categorized music data and log files. PMRS retrieves users’ historical data from log files and provides music recommendations across various genres. Evaluation of PMRS utilizes the million song dataset (MSD), alongside an examination of its performance using confidence score metrics across different music genres. To illustrate PMRS’s functionality, the authors developed a mobile application (Android version).

2.2.2 Recurrent neural networks. Convolutional neural networks used in music recommendation encounter some problems. We typically assume that data comes from some distribution, and all samples are independently and identically distributed (i.i.d.). However, in reality, user preferences for music are related to the order in which users listen to music. Incorporating temporal information about user music consumption into neural network learning greatly benefits deep learning-based music recommendation systems.

Reference^[16] utilized Recurrent Neural Networks (RNNs) based on user context and sequential preferences, using a large-scale music listening dataset from Spotify. They found that users’ recent consumption history and session-level contextual variables (such

as time of day or device type used) had strong predictive power for predicting which tracks users would listen to. This paper assumed characteristics of music such as short duration, repeat listening, session consumption, and context relevance, proposing a problem setting suitable for music recommendation. An effective method was proposed to utilize both user history and current context information, capable of generating real-time, personalized, and diverse recommendations. It outperformed existing methods in two evaluation tasks (session ranking and music ranking), demonstrating the importance of sequence and context information. Based on this observation, the authors proposed CoSeRNN. CoSeRNN represents user preferences as an embedding sequence, with one dedicated to each session. At the onset of a session, it anticipates a preference vector by analyzing past consumption history alongside current context. Subsequently, it employs an approximate nearest neighbor search algorithm in subsequent tasks to produce contextually relevant and timely recommendations.

Reference^[17] investigated the significance of song context and sequential order in automated music playlist generation and their role in recommending subsequent songs. Findings indicated a positive correlation between song context and the quality of next song recommendations, albeit potentially influenced by biases towards highly popular tracks. The study employed a technique introduced by Reference^[18], amalgamating a recurrent neural network (RNN) architecture featuring a single layer of gated recurrent units with a loss function tailored to enhance the ranking of next-item recommendations. During evaluation, given the current song within the playlist alongside all preceding songs, the RNN outputs a song score vector used for ranking the next song candidates.

Reference^[19] focused on music genre information, using the fma_medium dataset from the Free Music Archive, converting audio signal data into spectrogram images to serve as input for the neural network architecture. Two different architectures were employed: Convolutional Neural Networks (CNNs) and Convolutional Recurrent Neural Networks (CRNNs). CRNNs were utilized for both feature extraction and similarity distance computation to assess feature similarity. The recommendation system was developed by computing cosine similarity among extracted features from different songs. Upon acquiring fundamental music features, cosine similarity computation was extended to additional music features. CRNNs, incorporating considerations of both frequency characteristics and temporal patterns, exhibited superior overall performance. The authors also advocated for the inclusion of music genre information in music recommendation systems to enhance recommendation quality. The paper compared recommendation results considering music genre and not considering music genre, finding that users preferred recommendations considering music genre.

2.2.3 Attention mechanism and transformer. Although Recurrent Neural Networks (RNNs) can establish models for sequential relationships, they face challenges such as the vanishing and exploding gradient problems, especially with long sequences. Additionally, RNNs are not suitable for long sequences due to limited memory capacity and inability for parallel training. Attention mechanism, on the other hand, has the advantages of parallel computation and capturing long-range dependencies efficiently. The Transformer

model is entirely based on the attention mechanism, without any convolutional layers or recurrent neural network layers. While Transformers were initially introduced for sequence-to-sequence learning, they have now been applied in various modern deep learning realms, encompassing reinforcement learning, natural language processing, computer vision, and speech recognition.

Reference^[20] proposed the RecFormer architecture, which employs personalized temporal-aware transformers to simulate interactions between user histories within a single framework. Specifically, RecFormer adopts a masked language modeling task as the training objective to capture fine-grained track embeddings by reconstructing tracks. Additionally, it integrates temporal-aware self-attention mechanisms into the Transformer architecture, enabling the model to consider time-varying information between different users. RecFormer accurately and fairly recommends tracks for the EvalRS task based on users' personal listening habits. Linearized attention is utilized to reduce computational and memory complexity through kernel calculation. Studies on the proposed metrics' dissociation showcase the efficacy and equity of the suggested approach.

Reference^[21] merged user-song interactions with music chord progressions, leveraging music theory and attention mechanisms. Attention mechanisms were utilized to discern the significance of various segments within songs, thus examining the integration of music content into deep learning-driven music recommendation systems. The authors noted a recurring limitation in previous research where music content was typically inputted into recommendation models using fundamental audio features such as Mel-frequency cepstral coefficients. However, unlike tasks such as speech recognition, music audio is often comprised of multiple audio sources. As a result, low-level audio features rooted solely in the time or frequency domain may fail to capture the intricate nuances of music content, consequently hindering the performance of recommendation algorithms. To address this issue, they proposed a novel music recommendation framework that leverages chord representations and attention mechanisms. In their framework, rather than relying on basic audio features, music content is represented using chord structures. By adopting this alternative approach, they introduced a chord attention layer designed to discern users' nuanced preferences for various segments of music content. Through experimental evaluations conducted on a subset of the last.fm-1b dataset, they demonstrated the effectiveness of their approach. The study's findings highlight the potential of utilizing chord representations and attention mechanisms in enhancing the performance of music recommendation systems, shedding light on a promising direction for future research in the field.

Reference^[22] introduced a model called BERT4Rec, a sequential recommendation model based on bidirectional encoder representations of the Transformer. The task involves addressing the challenge of modeling the dynamic evolution of user preferences based on their historical behaviors—a crucial endeavor for recommendation systems. Traditional approaches have often employed sequential neural networks like recurrent neural networks (RNNs) to sequentially encode users' past interactions, generating hidden representations for recommendation. While proficient at capturing the unidirectional sequential patterns in user data, these methods

have limitations. In contrast, BERT (Bidirectional Encoder Representations from Transformers) represents a paradigm shift by training bidirectionally on user sequences, thereby enhancing recommendation outcomes. Leveraging the encoder component of the Transformer model, BERT utilizes attention mechanisms to grasp contextual relationships among users' historical interactions. Unlike traditional methods, BERT considers both past and future interactions simultaneously, leading to a more comprehensive understanding of user preferences. The proposed model integrates previous user interactions to inform bidirectional encoding, encompassing both left and right contexts concurrently. Experimental evaluations on benchmark datasets validate the model's efficacy, demonstrating its superiority over various state-of-the-art sequential models. This research significantly contributes to advancing recommendation systems by introducing a novel approach that effectively captures the evolving nature of user preferences, leading to more accurate and personalized recommendations.

2.2.4 Graph neural networks and graph-based models. Graph Neural Networks (GNNs) are frameworks that directly learn from graph-structured data using deep learning. In recent years, authors have utilized graph networks and graph models for music recommendation. For instance, in reference^[23], a new method for recommending music videos is introduced, which constructs a graph network capturing connections among music videos and users, while also encompassing multimodal attributes of the music videos. Through the fusion of diverse link prediction scores derived from local and global structures, a link prediction technique accounting for both local and global network architectures is developed to predict users' expected preferences for music videos, thus recommending music videos that users are likely to enjoy. Reference^[24] proposes a graph-based music recommendation framework that provides finer-grained music recommendations. The authors first discover recommendation clues from user ratings, referred to as user preference relationships, and then compute the likelihood of recommending specific songs. Experimental results indicate that the proposed framework outperforms two traditional algorithms, particularly in tackling the cold start issue in recommendations. In Reference^[25], utilizing social media data and music content, a dataset comprising tracks, albums, and artists is collected from [Last.fm](<http://last.fm/>), considering six object types and nine types of relationships within the aforementioned datasets. It was found that social media data is beneficial for music recommendation. The authors conducted a novel study where they integrated social media data with music auditory signals, reframing the recommendation conundrum as a ranking challenge on a unified hypergraph. They introduced a pioneering music recommendation algorithm on this hypergraph (MRH), adept at accurately capturing tagging relations, as illustrated in Figure 1. Evaluation of various recommendation algorithms using metrics such as precision, recall, F1 score, mean average precision (MAP), and normalized discounted cumulative gain (NDCG) revealed that the MRH algorithm consistently outperforms its counterparts. Notably, the superiority of MRH over Unified Graph Recommendation (RUG) underscores the efficacy of hypergraphs in modeling the intricate relationships within social media communities. This groundbreaking research sheds light on the potential of hypergraph-based approaches in

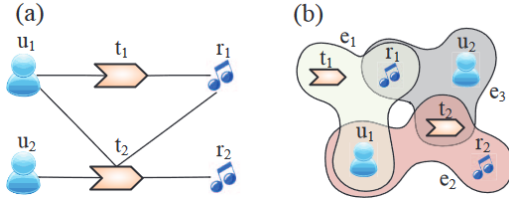


Figure 1: Tagging relations shown in two models: (a) ordinary graph model, and (b) unified hypergraph model.

addressing the complexity inherent in recommendation systems, particularly in scenarios where diverse data modalities, such as social media interactions and auditory signals, intersect. The findings not only contribute to advancing the field of recommendation algorithms but also pave the way for innovative solutions that leverage the synergies between different data sources for enhanced recommendation accuracy and relevance.

In Reference^[26], the authors obtain data about songs and users from the largest collection of free music functionality and metadata and propose the Hypergraph Embedding Music Recommendation framework (HEMR). This framework addresses the music recommendation problem by embedding and connecting all pertinent details concerning users, songs, artists, etc., via a hypergraph data structure. Initially, the information is structured and saved using a hypergraph-based format, which effectively represents complex multi-relational data. Subsequently, random walks and vertex embeddings are generated from the hypergraph data model. Through these embeddings, the system captures the rich information within the data, ultimately generating top K song recommendations for each user. Comprehensive experiments demonstrate that this framework and the hypergraph formulation outperform state-of-the-art benchmark methods in terms of quality, embedding representation, cold start experiments, and runtime. Notably, embedding the information from the hypergraph into a low-dimensional space proves to be an effective method for obtaining recommendations, including contextual and content information. This finding indicates that the music recommendation framework using hypergraph data structures and embedding methods outperforms existing benchmarks in multiple aspects, providing an efficient and scalable solution.

2.3 Personalized music recommendation system

2.3.1 Content-Based recommendation. Music, as an art form, encompasses rich information such as melody, rhythm, chords, instruments, and styles. Traditional collaborative filtering algorithms primarily rely on user-item interaction data for recommendations, often underutilizing the content information of items themselves. Therefore, extracting features from music audio and incorporating them into music recommendation systems can be highly beneficial.

One of the key tasks in content-based recommendation is representing the features of music, often done in vector form, primarily through handcrafted features or features extracted using deep learning. Handcrafted features include audio features such as rhythm, timbre, and chords, or metadata such as genres and styles tagged by users^[27],^[28].^[29] outlines commonly used music audio descriptors, literature^[13] utilizes convolutional neural networks to

extract embedding vectors of music features, and^[30] introduces a method to directly generate semantic representations of user music preferences from audio. This method starts by using a collection of given music tracks as indicators of user preferences and deduces high-level semantic descriptors for each track, thereby constructing a user model. Additionally, other studies^[31] have shown that machine learning can extract higher-level semantic descriptors such as genre, mood, and instruments from audio features, indicating that computationally derived music features can be used to assess individual music preferences.

Reference^[13] is a seminal work in the field of music recommendation, widely recognized for its pioneering use of deep learning techniques. The authors introduced a Convolutional Neural Network (CNN) approach that predicts latent variables of user-item interactions, derived from collaborative filtering data through matrix factorization. Notably, this CNN model uniquely extracts latent vector representations solely from audio data, demonstrating its versatility and effectiveness in utilizing auditory signals for improved recommendation outcomes. This influential study not only emphasizes the transformative potential of deep learning in music recommendation but also underscores the significance of incorporating diverse data modalities to understand user preferences better. The lasting impact of this research continues to shape content recommendation methodologies, inspiring subsequent studies and innovations in the field.

While the benefits of content-based recommendation include not needing to interact with other users' data, effectively avoiding cold-start problems and the long-tail effect, challenges such as high computational complexity and difficulty in audio feature extraction exist.

2.3.2 Context-Based recommendation. There are various interpretations of context concerning users, items, and their interactions, with different classifications of contexts available in literature^[32]. Here, we mainly discuss music recommendation systems based on user-related context, item-related context, and interaction-related context.

User-related context encompasses demographic details like gender, age, cultural background, and the activities users participate in. Item-related context includes playlists, listening history, and similar item attributes. Interaction-related context pertains to features linked with the listening experience, such as time and location. Firstly, concerning user-related context, Reference^[33] proposed several quantification measures to describe how closely aligned users' music preferences are with the mainstream and analyzed the differences in mainstream music between different countries. In rating prediction experiments, the proposed measures of mainstreamness were used to customize recommendations for users, resulting in an evaluation of the accuracy of each measure and an explanation of their strengths and weaknesses, demonstrating the applicability of these research findings to improving music recommendation systems.

Personalized tailored recommendations can improve the effectiveness of music recommendations for different audience groups, while item-based recommendations involve aspects such as playlists, listening histories, and similar factors. Reference^[34]

assumes that user characteristics impact the performance of algorithmic combinations and considers specific user groups determined by factors such as age, gender, country, and preferred music genres. Overall, the study finds that personalized tailored recommendations for different audience groups can enhance the effectiveness of music recommendations. Furthermore, item-based recommendation methods involve aspects such as playlists, listening histories, and other similar factors. Reference^[35] combined users' global preferences with contextual preferences, utilizing listening order, music metadata, and user-item matrices for recommendation, demonstrating better capability in handling sparse data. The authors leverage music listening records and corresponding metadata to learn embedding vectors of music pieces, effectively capturing the intrinsic features of music works. They obtain users' global and contextual preferences and recommend suitable music segments based on users' preferences. This study integrates users' global preferences with contextual preferences for recommendation, showing better capability in handling sparse data while effectively capturing the features of music works, providing personalized music recommendations.

Contextual information related to interactions in music recommendation is gaining more attention. With the rapid development of mobile devices enabling users to enjoy music anytime, anywhere, reference^[36] proposed location-based music recommendation, unveiling a novel venue-aware music recommendation system named VenueMusic, adept at identifying songs suitable for diverse types of popular venues encountered in everyday life. Through the utilization of Location-Aware Topic Modeling (LTM), the distinctive features of songs pertaining to specific venues were encoded within a latent semantic space, with venue feature vectors integrated into the same latent space as songs. To extract meaningful semantic representations of songs via LTM, a Music Concept Sequence Generation (MCSG) approach was developed.

2.3.3 Emotion-Based recommendation. Emotions significantly influence the motivation behind music consumption, as users use music to alter or maintain their emotions^[37]. The environment in which users listen to music greatly influences their music preferences, as music aids in mood regulation, self-awareness, and expression of social relationships^[38]. Emotions can be identified from music works, and various acoustic features of music can be employed to discern emotions in songs^[39]. Recognizing users' emotions can enhance the quality of music recommendations^[40].

Emotions significantly influence the motivation behind music consumption, as users utilize music to modulate or sustain their emotional states^[37]. The context of music listening profoundly shapes users' preferences, with music serving as a tool for mood regulation, self-reflection, and communication of social connections^[38]. Emotions are discernible within musical compositions, and various acoustic features can be leveraged to ascertain emotions embedded in songs^[39]. Understanding users' emotional states can enrich the precision of music recommendations^[40].

One approach to identifying users' emotions is to extract them from the text shared by users on their social networks. Reference^[41] employed lexicon-based sentiment analysis and machine learning-based sentiment analysis techniques to extract text from social network users and recommend music genres. The accuracy

of emotion classification using convolutional neural network algorithms for happiness, sadness, anger, fear, disgust, and surprise was higher than related works. Reference^[42] used social tags and emotions as the main features for recommending the next song in an autoplay playlist. The authors propose a strategy to recommend a set of tracks from a user's played song list as candidate songs for the next play, matching the main features of the played songs. To gather sufficient information and apply the recommendation strategy, the authors utilize third-party external information sources providing data about songs, including their popularity, emotions evoked by lyrics, low-level and high-level audio features, lyrics, etc. The system emphasizes the use of user-generated tags and emotional features to successfully predict the next important song played by the user. Another approach is to directly capture emotions from users' faces using a camera, although this may raise privacy concerns^[40], making it suitable for small-scale personal projects but unsuitable for large-scale commercialization.

Although emotion-based recommendation systems can improve recommendation effectiveness, they require large amounts of data and computation^[43]. Additionally, different individuals may perceive a song slightly differently, leading to ambiguity in the dataset.

2.3.4 Hybrid method-based recommendations. Recommendation systems based on hybrid methods utilize combinations of the recommendation model methods mentioned earlier. Compared to systems using only a single recommendation technique, such as systems combining content-based and collaborative techniques, they can recommend songs more effectively. These models can combine the strengths of each method for recommendation, and appropriate hybrid models can alleviate the weaknesses of each recommendation model. Reference^[44] introduces a novel hybrid recommendation system named EntreeC, which combines knowledge-based recommendation and collaborative filtering using cascade mixing for recommendation. The article discusses six hybrid techniques: weighted, mixed, switching, feature combination, feature augmentation, and meta-level. The authors also demonstrate how semantic ratings obtained by the system's knowledge-based component can enhance the effectiveness of collaborative filtering.

2.3.5 Reinforcement learning-based music recommendation systems. Traditionally, recommendation systems have treated recommendation problems as classification or prediction tasks. However, there is now widespread recognition that modeling recommendation problems as sequential decision-making problems can better reflect the interaction between users and the system. Although reinforcement learning has been used in the field of recommendation for decades, it has not been practical in practice. However, since the introduction of deep reinforcement learning, applying reinforcement learning to recommendation problems with large-scale state and action spaces has become feasible. The modeling of recommendation systems can shift from traditional classification or prediction problems to sequential decision-making problems to better reflect the interaction between users and the system. The introduction of deep reinforcement learning makes it feasible to apply reinforcement learning to recommendation problems with large-scale state and action spaces.

Both collaborative filtering and content-based recommendation methods encounter challenges such as the cold start problem and recommendation diversity issues. Addressing these concerns, reference^[45] presents the Reinforcement Personalized Music Recommendation System (RPMRS) designed to alleviate these obstacles. RPMRS comprises two main components: utilizing WaveNet and Word2Vec models to extract deep representations of audio and lyrics, respectively, followed by a novel content-based recommendation approach. Additionally, reinforcement learning is employed to understand user preferences from song playback logs, allowing the system to refine its recommendations continually. Reinforcement-based recommendation systems have the advantage of adapting their strategies based on user feedback, aiming to maximize cumulative user rewards. RPMRS operates dynamically, constantly optimizing to adjust recommendation lists according to evolving user preferences. Experimental results confirm the superiority of hybrid features over solely audio-based or lyric-based ones for content recommendation. Furthermore, the integration of reinforcement learning leads to dynamic and more personalized music recommendations, showcasing the system's adaptability to individual user preferences over time. This groundbreaking research not only tackles prevalent challenges in music recommendation but also highlights the potential of reinforcement learning techniques in crafting responsive and personalized recommendation systems across various domains. Reference^[46] presents PHRR (Personalized Hybrid Music Recommendation based on Reinforcement Learning), an innovative algorithm aimed at enhancing personalized music recommendation by tailoring song sequences to individual audience preferences. By embracing hybrid recommendation concepts, PHRR seamlessly integrates Weighted Matrix Factorization (WMF) and Convolutional Neural Network (CNN) models rooted in content and collaborative filtering methodologies, enabling effective learning and prediction of music features to enrich the recommendation process. PHRR conceptualizes the reinforcement learning problem as a Markov Decision Process, leveraging audience preferences for individual songs and song transitions to model the recommendation environment. This approach simulates audience decision-making behaviors, enhancing recommendation fidelity. Furthermore, the model continuously updates itself based on evolving audience preferences, ensuring adaptability and relevance over time. This innovative approach not only combines the strengths of content-based and collaborative filtering techniques but also harnesses reinforcement learning to deliver personalized and contextually relevant music recommendations. By simulating audience interaction processes and refining recommendations iteratively based on real-time feedback, PHRR represents a significant advancement in personalized music recommendation algorithms. The PHRR algorithm proposed in reference^[46] combines models based on content and collaborative filtering and utilizes reinforcement learning to simulate audience decision-making processes, thereby achieving more personalized music recommendations that better match audience preferences. Reference^[47] addresses a prevalent issue in current music recommendation systems, where recommendations often prioritize the highest-rated songs in a myopic, greedy manner, lacking foresight and long-term capture of user preferences. In response, the

authors propose a novel methodology inspired by the exploration-exploitation trade-off in reinforcement learning, particularly the multi-armed bandit problem. The exploration-exploitation trade-off in music recommendation involves balancing between recommending familiar songs likely to be enjoyed by the user (exploitation) and suggesting new or less popular songs to gather more information about user preferences (exploration). By framing recommendation as a multi-armed bandit problem, the authors employ bandit methods to address the cold-start problem and enhance recommendation performance. To discern user preferences, a Bayesian model is utilized, considering both audio content and recommendation novelty. This Bayesian approach allows the system to adaptively update its understanding of user preferences based on observed interactions. Additionally, to expedite Bayesian inference, piecewise linear approximation models and variational inference algorithms are employed, ensuring efficient and scalable computation of user preferences. An advantage of this approach lies in its integration of a unified model for both music recommendation and playlist generation. By treating recommendation and playlist generation as complementary tasks within the same framework, the system can offer cohesive and personalized music experiences for users, seamlessly transitioning between recommending individual songs and curating playlists tailored to their preferences. This innovative methodology represents a significant advancement in music recommendation systems, offering a principled approach to balancing exploration and exploitation, leveraging Bayesian inference for personalized recommendations, and unifying recommendation and playlist generation within a single framework. The method proposed in reference^[47] addresses the greedy problem in music recommendation by balancing exploration and exploitation, using multi-armed bandit methods, and incorporates Bayesian models to consider audio content and recommendation novelty, thus improving recommendation performance. Reference^[48] proposed a novel music recommendation strategy, NRRS (Non-Intrusive Perceptual and Reinforcement Learning-based Recommender System). The authors developed a novel recommendation framework based on wireless sensing and reinforcement learning, which perceives, learns, and adjusts to users' current preferences in real-time during music listening sessions. The established music recommendation prototype monitors individual physiological features and captures song characteristics and dynamic preferences, thereby providing users with a better music listening experience. The NRRS strategy proposed in reference^[48] utilizes reinforcement learning and wireless sensing to dynamically adapt to users' current music preferences, providing a better music recommendation experience by monitoring individual physiological features and capturing music characteristics.

For further information on the application of reinforcement learning in recommendation systems, please refer to reference^[49].

3 EVALUATION METHODS

Evaluation metrics for recommendation systems have their roots in the domains of machine learning and information retrieval. These metrics serve as vital tools for assessing the effectiveness and performance of recommendation algorithms. Reference^[50] categorizes evaluation methods for recommendation systems into three main

types: offline testing, online testing, and user-based subjective evaluation. Offline evaluation methods rely on existing datasets of user-item interactions and do not require user participation during the evaluation process. The following mainly discusses the application of offline evaluation, online evaluation, and user-based subjective evaluation in music recommendation systems.

Offline evaluation in recommendation systems utilizes existing datasets and doesn't require user participation during assessment. Common performance metrics include root mean squared error (RMSE), recall, precision, average precision (AP), mean average precision (mAP), normalized discounted cumulative gain (NDCG), and mean percentile rank. Mean Average Precision (MAP) specifically measures the average proportion of relevant items in the recommended list by the system; Normalized Discounted Cumulative Gain (NDCG) is used to measure the quality of recommendation results, especially in ranking tasks, which involve sorting items according to the likelihood of being liked by the user; Mean Percentile Rank (MPR) is used to measure the average position of recommended items in the user's ranking. Specifically, it measures the percentile ranking of items actually clicked or interacted with by the user in the recommendation list. For more details on the calculation of these metrics, refer to reference^[51]. Below are several methods used for calculating metrics:

(1) Mean Squared Error, as shown in the following formula:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (1)$$

Where, n is the total number of samples, y_i is the actual value of the i -th sample, $\hat{y}_i$ is the predicted value of the i -th sample.

(2) Recall, as shown in the following formula:

$$Recall = \frac{TP}{TP + FN} \quad (2)$$

Where, TP represents True Positive and FN represents False Negative.

(3) Precision, as shown in the following formula:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

where TP represents True Positives, TN represents True Negatives, FP represents False Positives, and FN represents False Negatives.

(4) Average Precision (AP), as shown in the following formula:

Assuming the ranking results of a recommendation system for a certain user's test set are as follows: The true labels for the recommendation sequence $N=1, 2, 3, 4, 5, 6$ are 1, 0, 0, 1, 1, 1, respectively. Then, for the above sequence, precision@N is calculated as 1/1, 1/2, 1/3, 2/4, 3/5, 4/6.

AP (Average Precision) is calculated by summing up the precisions of positive samples in precision@N, i.e., $AP = (1/1 + 2/4 + 3/5 + 4/6) = 0.6917$.

(5) Normalized Discounted Cumulative Gain (NDCG): as shown in the following formula:

$$NDCG@k = \frac{DCG@k}{IDCG@k} \quad (4)$$

Where DCG@k is the Discounted Cumulative Gain at the top k recommended results, and IDCG@k is the Ideal Discounted Cumulative Gain at the top k recommended results.

(6) Mean Percentile Rank (MPR): as shown in the following formula:

$$MPR = \frac{1}{N} \sum_{i=1}^N \frac{r_i}{n_i} \quad (5)$$

Where: N is the total number of users or instances, r_i is the rank of the relevant item for the user or instance, and n_i is the total number of items for the user or instance.

There are also some other types of metrics, such as novelty, diversity, familiarity, popularity, serendipity, etc. However, these metrics are difficult to describe mathematically, such as measuring serendipity, and these metrics may have multiple different measurement methods. Offline evaluation methods are still the main approach for researchers because academia lacks interaction with a large number of real users. Although offline evaluation has its advantages, the accuracy of recommendations is not always correlated with user satisfaction^[52]. The key issue in offline research is being overly confident in the results, as offline evaluation results often exhibit low variance and high bias, the latter of which may lead to unreasonable confidence in the evaluation results^[53]. Furthermore, if the goal of the recommendation system is exploratory, this evaluation method does not take into account the familiarity of the audience with the recommended items.

Online evaluation is widely employed in industry due to the vast user base, enabling direct user involvement in assessments. A prevalent method for online evaluation is A/B testing (also known as split testing and bucket testing), which aims to compare the effectiveness of two versions of content to identify which version is more appealing to visitors/viewers^[54]. Reference^[55] used A/B testing to evaluate a new recommendation algorithm, Bart, in a live music recommendation service. In A/B testing, the authors randomly divided the production traffic into two groups, one using the Bart algorithm for recommendations and the other using a baseline algorithm called Control, and then compared the performance of the two groups. To avoid user confusion, the authors limited the exploration rate to within 10% and set the causal unit as each recommendation for a day. The results showed that the Bart algorithm performed better than the baseline algorithm when compared in the A/B test, especially when considering interactions. However, in online testing, the performance of the Bart algorithm may differ from offline experiments due to the breaking of the independence assumption of rewards, making it difficult to accurately measure performance differences. Although the results of online testing may slightly differ from offline experiments, they still indicate that the Bart algorithm can provide significant improvements, thus demonstrating its practicality in real-world production environments.

User subjective evaluation of recommendation systems has obvious advantages compared to online and offline testing, as it can deeply reveal the intrinsic features of user experience. However, such evaluations are rare because it is difficult to collect enough participants to draw meaningful conclusions.

User-centric subjective evaluation methods can investigate user experience, including user engagement and satisfaction. Reference^[56] introduces ResQue (Recommendation System User Experience Quality), a holistic evaluation framework designed to assess multiple facets of recommendation systems. ResQue goes beyond

merely evaluating the quality of recommended items; it also encompasses the usability of the system, its overall usefulness to the user, the quality of the interface and interaction design, user satisfaction, and the subsequent impact of these factors on user behavioral intentions. ResQue aims to provide a comprehensive understanding of the user experience within recommendation systems by examining not only the effectiveness of the recommendations but also the user's interaction with the system as a whole. This includes evaluating how intuitive and easy-to-use the system interface is, the relevance and usefulness of the recommendations provided, and ultimately, the degree to which users are satisfied with their experience. Additionally, ResQue explores the downstream effects of these factors on user behavior, such as their likelihood to purchase recommended products or to return to the system for future interactions. By considering a wide range of dimensions that contribute to the overall user experience, ResQue offers a robust framework for researchers and practitioners to evaluate and improve recommendation systems comprehensively. This multi-dimensional approach enables a more nuanced understanding of user satisfaction and behavior, ultimately leading to the development of more effective and user-centric recommendation systems. The article also demonstrates the application of psychometric methods to data collected from large-scale user surveys to validate the comprehensive standards. Validation results support the consistency, validity, and reliability of the selected standards and explain the key determinants of user experience quality and adoption of recommendation technologies. The ResQue framework proposed in reference^[56] validates the consistency, validity, and reliability of comprehensive standards through psychometric methods, explaining the key factors influencing user experience quality and adoption of recommendation technologies. Reference^[57] emphasizes that while recommendation algorithm accuracy is crucial, it represents just one aspect of the broader user experience in recommendation systems. To address this, the paper introduces a novel framework that adopts a user-centric approach to comprehensively evaluate recommendation systems. This framework surpasses traditional metrics by connecting objective system aspects with user behaviors through subjective system aspects and experiences. The framework goes beyond quantitative system performance to include qualitative dimensions of user perception and interaction. By integrating subjective system aspects and experiences, it aims to capture nuanced aspects of user satisfaction, engagement, and overall experience with the recommendation system. Additionally, the framework recognizes the influence of individual and contextual characteristics on user experience. By considering factors like user preferences, demographics, and situational contexts, it aims to provide a nuanced understanding of how various users perceive and interact with recommendation systems. The framework proposed by reference^[57] comprehensively evaluates recommendation systems through a user-centric approach, linking objective system aspects with objective user behaviors and considering the influence of individual and contextual characteristics on user experience. Reference^[58] meticulously examines user control in context-aware music recommendation systems, presenting ContextPlay as a system allowing users simultaneous control over contextual features and music preferences. Through a detailed mixed-design study, the

paper explores the effects of controlling contextual features on four key aspects of music recommendation systems: perceived quality, diversity, effectiveness, and cognitive load. Compared to baseline systems limiting user control to music preferences alone, the addition of control over contextual features significantly improves users' perception of recommendation quality without increasing cognitive load. The study highlights the importance of contextual elements like mood, weather, and location in shaping users' views of recommendation systems, emphasizing their substantial impact on the overall user experience. Furthermore, the research reveals intriguing insights into user behavior, showing a heightened tendency among users to adjust both contextual and personal profiles during relaxation activities. This finding underscores the dynamic relationship between user preferences and contextual factors, emphasizing the importance of incorporating user control mechanisms to accommodate these nuances in context-aware recommendation systems. The study provides valuable insights into the role of user control in enhancing the user experience in context-aware music recommendation systems, offering practical implications for designing and implementing such systems to better align with user preferences and contextual nuances.

4 DATASET

There are numerous publicly available datasets for music recommender systems, each containing different information, thus researchers have different focuses. The Million Song Dataset^[59] includes information on 1 million songs, totaling approximately 280GB, providing rich features for music recommendation-related competitions. The Music Streaming Sessions Dataset (MSSD)^[60] is Spotify's dataset, containing data from 160 million user interaction sessions and song features during interactions. Initially released to support the Spotify Sequential Skip Prediction Challenge of the WSDM Cup Challenge 2019, the task is to predict user playback behavior based on the labels of the first half of the playback sequence (skip=1, play=0). The Last.fm 1K dataset^[61] collects the complete listening history of 992 users from July 2005 to May 2009, comprising about 20 million records. It includes two tables: the user's historical interaction records table, including userID, event timestamp, artistID, artist_name, songID, and song_name, and the user feature table, including gender, age, country, registration time, etc., and provides information on the timestamp of listening events, which can be used for session-based or streaming recommendation tasks. The Last.fm 360K dataset^[61], also provided by Last.fm, contains the same user information as Last.fm 1K but has 360K users and collects records of user plays of artists (user, artist, play_count). The Yahoo! Music dataset^[62] provides a large-scale user-item rating dataset, with different levels of ratings (tracks, albums, artists, and genres) and extremely high sparsity (99.96%), making the task particularly challenging and suitable for rating prediction tasks to distinguish liked songs from others. The #nowplaying dataset^[63] collects 49 million user listening event data from Twitter, including 144,011 artists, 1,346,203 songs, and 4,150,615 users, with fields such as userid, source_of_the_tweet, timestamp, song_name, MBID, etc., suitable for music recommendations related to social networks. The #nowplaying-RS dataset^[64], also collected from Twitter, provides rich song information features and user context features compared

to #nowplaying. Additionally, its major feature is user emotional tags, which can be used in music sentiment analysis and exploring the emotional state behind user listening events. The Million Musical Tweets Dataset (MMTD) [65] is another listening event dataset collected from Twitter, characterized by including specific latitude and longitude geographic information, with a broad distribution of listening events across countries, but with uneven geographic distribution, suitable for supporting context-related music recommendation tasks. The NetEase Cloud Music Data (NCM) dataset [4] comes from NetEase Cloud Music and includes over 57 million music content cards recommended to randomly selected 2,085,533 users from November 1, 2019, to November 30, 2019. It also provides interaction behavior data at the user and playlist levels, suitable for bundled recommendation. The 30Music dataset [66] is collected from lastfm's API, providing user-rated liking score data for distinguishing liked songs from others. The LFM-1b dataset [67], provided by Last.fm, encompasses over 100 million music listening events recorded by more than 120,000 users. Each event is meticulously identified by artist, album, and track names, accompanied by timestamps. This rich dataset is highly suitable for tasks such as music retrieval and recommendation, as well as for in-depth exploration of user music preferences and consumption behaviors. Building upon this, the LFM-1b User Genre Profile dataset [68] extends the LFM-1b dataset by offering detailed information on the music genre preferences of over 120,000 listeners. This extension is intricately associated with the original LFM-1b dataset, providing a comprehensive resource for analyzing genre-specific listening habits and further enhancing the understanding of user preferences in the context of music consumption. Each user creates a preferred genre vector to represent the genres they like. This dataset can be used to explore the correlation between genres and user music preferences. Spotify Playlists dataset [69] were created to improve collaborative filtering recommendation systems by addressing the challenge of collecting context information and can be used for context-based music recommendation. The MusicMicro dataset [70] considers methods of retrieval that are very popular due to cultural differences in music perception and consumption. To bridge this gap, music-related Weibo posts are first identified, from which information about location and music items is extracted, annotated for each item, and the results of the spatiotemporal music retrieval method are analyzed to explore the relationship between music preferences and spatial attributes as well as the temporal stability of listening patterns. The Melon Playlist Dataset [71] is a publicly accessible resource containing Melon spectrograms. This extensive dataset comprises 649,091 songs and 148,826 related playlists, annotated with 30,652 unique tags. All data originates from Melon, a leading South Korean streaming service, making it especially useful for music information retrieval tasks, including automatic tagging and playlist continuation.

5 CHALLENGES

Music recommender systems encounter diverse challenges, notably encompassing the cold start issue, fairness concerns in recommender systems, and the interpretability challenge in recommender systems.

The cold start problem poses a notable hurdle in the recommendation sphere, especially in the field of music recommendations. When new users register, the system lacks historical data for recommendations. In such cases, the recommender system cannot make user-based or item-based recommendations to new users. Reference [72] proposes a solution to the cold start problem based on community detection algorithms. The method calculates the similarity between users and items by projecting a bipartite network and then uses the Louvain algorithm to detect communities in the projected unimodal network. New records are updated to the original community, and music recommendations are made for user groups accordingly. Reference [73] discusses addressing the cold start and long-tail problems in music recommendation. The authors utilize convolutional neural networks (CNNs) to automatically extract music content and contextual features, transforming them into representations in the latent space to generate music recommendations. Finally, a user-centric evaluation is conducted to compare the performance of the recommendation system from the users' perspective. Reference [74] explores the potential of music recommendation based on audio content. This approach enables even new artists with fewer listeners to be recommended. The study's findings indicate that the content-based method provides a more equitable opportunity for music exposure for all artists, regardless of their fame and popularity.

The widespread application of machine learning-based recommender systems has brought about algorithmic bias issues. To address fairness issues in the music recommendation domain, there is a need to establish definitions of fairness, such as fairness for whom, considering that stakeholders in the music realm encompass users, composers, artists, music streaming service providers, among others. Overcoming the shortcomings of evaluation metrics and optimizing objectives and dataset biases are necessary to improve fairness. Reference [75] collects prominent fairness definitions in algorithm classification problems, explains the principles behind these definitions, and demonstrates each definition in a unified case study. The authors analyze why the same case may be considered fair according to certain definitions but unfair according to others. Reference [76] investigates the fairness in collaborative filtering recommendation systems, emphasizing the sensitivity of such systems to biases present in historical data, which may lead to unfair predictions for minority user groups. The authors identify shortcomings in existing fairness measures and propose four new metrics to address various forms of unfairness.

Addressing the interpretability problem in research on music recommendations can improve both the algorithms and user experience. Providing reasons for recommending music to users can enhance user engagement and trust. Reference [77] explores the impact of personal characteristics on the effectiveness of explanations in music recommendation systems. The study involves the development of an explanatory music recommender system and the implementation of online studies using an internal object design methodology. The findings indicate that personal traits significantly influence users' perceptions and understanding of music recommendations. This research highlights the critical importance of incorporating individual differences into the design of explanatory

systems. By tailoring explanations to align with personal characteristics, the study demonstrates the potential to enhance user satisfaction and engagement with music recommendation systems. These insights suggest that a personalized approach to explanation could lead to more effective and user-friendly recommendation experiences, ultimately fostering greater user trust and loyalty. The findings suggest that personal characteristics exert a noteworthy influence on the interaction and perception of recommender systems, and this impact varies with the addition of explanations. Specifically, individuals with low cognitive demands benefit from explanatory suggestions. For those with high cognitive demands, the authors observe that explanations can reduce their confidence. Reference^[78] discusses how to address interpretability issues in the context of music recommendation systems, including the purpose, dimensions, methods, evaluation, and challenges of interpretability. The authors argue that interpretability plays a crucial role in enhancing user trust, satisfaction, and loyalty to music recommendations, especially when the recommendations deviate from users' expectations or are inappropriate. They review general concepts of interpretable artificial intelligence and discuss their applicability to the specific scenarios of music recommendations. Additionally, they demonstrate how to integrate interpretability components into music recommendation systems and provide examples of different forms of recommendation explanations. Furthermore, the authors discuss the requirements and strategies for evaluating the quality of music recommendation explanations and the current challenges and research prospects for introducing interpretability into large-scale industrial music recommendation systems.

6 CONCLUSION AND TRENDS

With the exponential growth of online information, individuals find themselves grappling with an overwhelming influx of data, a phenomenon commonly referred to as information overload. Consequently, many turn to recommendation systems as a means of navigating this deluge, often without a clear understanding of the intricate technologies driving these recommendations. Information overload refers to the situation where individuals face an overwhelming amount of information within a short period, making it challenging to process effectively. This phenomenon arises from several factors. Firstly, the internet and social media have led to an explosion of available information, inundating individuals with data. The accessibility of information via smart devices further increases the volume and frequency of incoming data. Limited filtering and processing capabilities exacerbate the issue, hindering the differentiation between relevant and irrelevant information. Redundant information and repetitive content also contribute to cognitive overload. Interruptions from advertisements and irrelevant content, such as pop-up ads, divert attention away from important tasks. Multitasking exacerbates the problem as individuals attempt to juggle multiple tasks simultaneously. The rapid evolution of digital platforms and tools adds complexity to information management. Social pressures from peers, family, and colleagues also contribute to the information load. In conclusion, information overload is a pervasive issue in contemporary society, necessitating strategies such as information filtering, time management, and improved digital literacy to mitigate its effects. By harnessing algorithms that

prioritize relevance and user preferences, recommendation systems can effectively streamline the information consumption process, making it more manageable and enhancing user satisfaction. Thus, integrating recommendation systems into broader efforts to address information overload represents a necessary and beneficial direction for mitigating this pervasive issue in modern society.

Presently, recommendation systems grapple with a myriad of obstacles, including the vexing cold start problem, equity concerns, and the imperative for interpretability, thereby demanding deeper scrutiny and investigation from the academic community. In essence, while recommendation systems undeniably enhance convenience in our daily lives, they concurrently encounter novel challenges such as the cold start dilemma, equity considerations, and the need for interpretability, thus underscoring the necessity for continued research and exploration in this field. The cold start problem in recommendation systems encompasses the challenges encountered when dealing with new users or new items, which are categorized into user cold start and item cold start. User cold start arises from the absence of historical behavioral data for new users, impeding the system's ability to accurately comprehend their preferences and interests, consequently diminishing recommendation accuracy. Conversely, item cold start pertains to the difficulty of recommending new items without sufficient historical data to assess their characteristics and relevance accurately. This issue arises from both new users joining the system, resulting in a lack of their historical behavioral data, and new items lacking historical data for the system to evaluate. The cold start problem arises from the system's deficiency in historical data for new users or items, leading to a reduction in recommendation accuracy. Addressing this challenge necessitates the development of innovative approaches that enable recommendation systems to effectively adapt to the presence of new users and items, thereby enhancing their ability to provide accurate and personalized recommendations in diverse contexts; The fairness concern in recommendation systems centers on preventing recommendations from unfairly favoring users based on personal characteristics, covering individual fairness and group fairness. Individual fairness aims to offer equitable recommendations to each user without bias towards specific user groups. Conversely, group fairness strives for fair recommendations across diverse groups, reducing biases towards demographics like gender, race, or age. This issue arises from potential biases in how the system treats users or groups, which could stem from relying too heavily on certain user data or design biases favoring specific groups. Ensuring fairness is crucial to mitigate biases and ensure equitable recommendations for all users or groups. By prioritizing fairness, recommendation systems can uphold principles of equality and inclusivity, building trust and satisfaction among users while promoting diversity and representation in recommendations; The challenge of interpretability in recommendation systems involves elucidating the rationale behind recommendation decisions, encompassing explanation and transparency. Recommendation explanation enables users to understand why specific recommendations are made and the underlying logic of the system. Recommendation transparency ensures that the decision-making process is clear, allowing users to understand how their data is used. This challenge arises from the lack of transparency in the recommendation process, leading to users being

unable to understand recommendation outcomes. Complex algorithms exacerbate this issue. Addressing interpretability is crucial to build user trust and engagement. By prioritizing interpretability, recommendation systems empower users to make informed decisions and understand recommendations. This transparency enhances satisfaction and promotes accountability and ethics in algorithm development.

This paper focuses primarily on music recommendation systems, concentrating on the developments in the field of music recommendation over the past decade. From two different perspectives, namely the structure of neural networks and personalized recommendation systems, we discuss the principles, characteristics, and applications of various models, as well as list some commonly used datasets in the field of music recommendation. We also review the main evaluation methods used in music recommendation systems and the challenges they face, as well as introduce possible future research directions in music recommendation.

Multi-modal music recommendation systems: The world is multi-modal, and humans observe it through various senses such as vision, hearing, smell, taste, and touch. While listening to music primarily involves hearing, text-based retrieval is mainly used in music searches. Reference^[79] trained an artificial intelligence chatbot for suggesting personalized songs according to users' moods. The outcomes illustrated the model's proficient learning ability and its success in offering music recommendations aligned with users' moods.

Generative Artificial Intelligence Combined with Music Recommendation: The combination of large language models (LLMs) from natural language processing with recommendation systems is one of the recent research directions of scholars, and users may be able to achieve better recommendations by issuing commands in natural language^[80]. Recently, music generation has received significant attention from researchers, as referenced in [81], [82]. Generative music and music recommendation systems are also recent areas of interest for researchers, utilizing user text prompts to generate personalized music to form personalized music recommendations^[83].

We hope this review provides convenience for researchers studying music recommendation systems, offering unresolved challenges and the latest research directions for future exploration.

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